# Chapter 10. Separation Theorems in the Plane

Jordan separation, invariance of domain, Jordan curve theorem, graphs in the plane, winding numbers

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## 10.1 The Jordan separation theorem

**Motivation.** Fundamental groups show that a simple closed curve disconnects the plane; we start with this easier half.

Open subsets of $S^2$ and $\mathbb{R}^2$ are locally path connected, so their components are open and equal their path components. $A$ **separates** a connected $X$ if $X - A$ is not connected (into $n$ components if $X - A$ has $n$ components). An **arc** is a space homeomorphic to $[0,1]$; a **simple closed curve** one homeomorphic to $S^1$.

**Lemma 10.1 (61.1)**

Let $C \subset S^2$ be compact, $b \in S^2 - C$, and $h : S^2 - b \to \mathbb{R}^2$ a homeomorphism. If $U$ is a component of $S^2 - C$ not containing $b$, then $h(U)$ is a bounded component of $\mathbb{R}^2 - h(C)$; if $U$ contains $b$, then $h(U - b)$ is the unbounded component. In particular $S^2 - C$ and $\mathbb{R}^2 - h(C)$ have the same number of components.

**Proof.** $U - b$ is connected for every component $U$: if $b \in U$ and $A, B$ separate $U - b$, take a disc neighbourhood $W \subset U$ of $b$ missing $C$; the connected $W - b$ lies in $A$, say, so $b$ is not a limit point of $B$ and $A \cup \{b\}$, $B$ separate $U$. Hence the $h(U_\alpha - b)$, $U_\alpha$ the components of $S^2 - C$, are disjoint connected open sets with union $\mathbb{R}^2 - h(C)$, its components. Extend $h$ to $H : S^2 \to \mathbb{R}^2 \cup \{\infty\}$, $H(b) = \infty$. For $U \ni b$, $H(U)$ is a neighbourhood of $\infty$, so $h(U - b)$ contains the complement of a compact set: it is unbounded, and the other components are bounded. ■

**Lemma 10.2 (61.2, nulhomotopy lemma)**

Let $a, b \in S^2$, $A$ compact and $f : A \to S^2 - a - b$ continuous. If $a$ and $b$ lie in the same component of $S^2 - f(A)$, then $f$ is nulhomotopic.

**Proof.** With $S^2 = \mathbb{R}^2 \cup \{\infty\}$, $a = 0$, $b = \infty$: if $g : A \to \mathbb{R}^2 - 0$ and 0 is in the unbounded component of $\mathbb{R}^2 - g(A)$, then $g$ is nulhomotopic. Take a ball $B \ni 0$ containing $g(A)$, $p \notin B$, and a path $\alpha$ in $\mathbb{R}^2 - g(A)$ from 0 to $p$. $G(x,t) = g(x) - \alpha(t)$ is a homotopy in $\mathbb{R}^2 - 0$ from $g$ to $k(x) = g(x) - p$, and $H(x,t) = tg(x) - p$ one from the constant $-p$ to $k$ ($tg(x) \in B$, $p \notin B$). ■

**Theorem 10.3 (61.3, Jordan separation theorem)**

A simple closed curve $C$ in $S^2$ separates $S^2$.

**Proof.** Suppose $S^2 - C$ is connected, hence path connected. Write $C = A_1 \cup A_2$, arcs meeting in their end points $a, b$; $X = S^2 - a - b$, $U = S^2 - A_1$, $V = S^2 - A_2$. Then $X = U \cup V$, $U \cap V = S^2 - C$ path connected. If $i_* : \pi_1(U, x_0) \to \pi_1(X, x_0)$ and similarly $j_*$ are trivial, $\pi_1(X, x_0) = 1$ (Theorem 9.38), whereas $X \cong \mathbb{R}^2 - 0$. A loop $f$ in $U$ at $x_0$ induces $h : S^1 \to U$, $h \circ p_0 = f$. $h(S^1)$ misses the connected $A_1 \ni a, b$, so $i \circ h$ is nulhomotopic (Lemma 10.2), $(i \circ h)_*$ is trivial (Lemma 9.20), and $i_*[f] = (i \circ h)_*[p_0]$. ■

**Theorem 10.4 (61.4, general separation theorem)**

If $A_1, A_2$ are closed connected subsets of $S^2$ with $A_1 \cap A_2 = \{a, b\}$, then $C = A_1 \cup A_2$ separates $S^2$.

**Proof.** $C \neq S^2$: $S^2 - a - b$ is connected, $C - a - b$ is separated by the $A_i - a - b$. The proof of Theorem 10.3 used only that $A_1, A_2$ are closed, connected and meet in $\{a, b\}$. ■

## 10.2 Invariance of domain

**Motivation.** A continuous injection of an open $U \subseteq \mathbb{R}^2$ into $\mathbb{R}^2$ has open image.

**Lemma 10.5 (62.1, homotopy extension lemma)**

Let $X \times I$ be normal, $A \subset X$ closed, and $f : A \to Y$ continuous with $Y$ open in $\mathbb{R}^n$. If $f$ is nulhomotopic, it extends to a continuous nulhomotopic $g : X \to Y$.

**Proof.** Let $F : A \times I \to Y$ be a homotopy from $f$ to the constant $y_0$; set $F(x, 1) = y_0$ on $X$. $F$ is continuous on the closed $(A \times I) \cup (X \times 1)$, so extends (Tietze, coordinatewise) to $G : X \times I \to \mathbb{R}^n$. $U = G^{-1}(Y)$ is open and contains $(A \times I) \cup (X \times 1)$; by the tube lemma there is an open $W \supset A$ with $W \times I \subset U$. Urysohn ($X \cong X \times 0$ normal) gives $\phi : X \to [0, 1]$, $\phi = 0$ on $A$, $\phi = 1$ off $W$. Then $(x, \phi(x)) \in U$, so $g(x) = G(x, \phi(x))$ maps $X$ into $Y$ and extends $f$, and $H(x, t) = G(x, (1-t)\phi(x)+t)$ stays in $Y$ and joins $g$ to $y_0$. ■

**Lemma 10.6 (62.2, Borsuk lemma)**

Let $a, b \in S^2$, $A$ compact, and $f : A \to S^2 - a - b$ continuous and injective. If $f$ is nulhomotopic, then $a$ and $b$ lie in the same component of $S^2 - f(A)$.

**Proof.** $f(A) \cong A$, so take $f$ an inclusion, $a = 0$, $b = \infty$: if $A \subset \mathbb{R}^2 - 0$ is compact and $j : A \to \mathbb{R}^2 - 0$ is nulhomotopic, 0 is in the unbounded component of $\mathbb{R}^2 - A$. Suppose the component $C \ni 0$ is bounded, $D$ the union of the others (both open). $C \cup A = \mathbb{R}^2 - D$ is compact, and Lemma 10.5 extends $j$ to $k : C \cup A \to \mathbb{R}^2 - 0$. Let $h = k$ on $C \cup A$, $h(x) = x$ on $D \cup A$; $h : \mathbb{R}^2 \to \mathbb{R}^2 - 0$ is continuous (pasting lemma). For a closed ball $B$ of radius $M$ with $C \cup A \subset \operatorname{Int} B$, $h$ is the identity on $\operatorname{Bd} B$, so $x \mapsto Mh(x)/\|h(x)\|$ retracts $B$ onto $\operatorname{Bd} B$, contradicting Theorem 9.19. ■

**Theorem 10.7 (62.3, invariance of domain)**

If $U \subset \mathbb{R}^2$ is open and $f : U \to \mathbb{R}^2$ is continuous and injective, then $f(U)$ is open and $f^{-1} : f(U) \to U$ is continuous.

**Proof.** It suffices that $f$ is an open map into $S^2 \supset \mathbb{R}^2$.

**Step 1.** For a closed ball $B \subset U$, $f(B)$ does not separate $S^2$: for $a, b \notin f(B)$, $f|B : B \to S^2 - a - b$ is nulhomotopic ($B$ contractible) and injective; apply Lemma 10.6.

**Step 2.** $f(\operatorname{Int} B)$ is open. $C = f(\operatorname{Bd} B)$ separates $S^2$ (Theorem 10.3). Let $V$ be the component of $S^2 - C$ containing the connected $f(\operatorname{Int} B)$, $W \neq \emptyset$ the union of the others. If $a \in V - f(\operatorname{Int} B)$, then $a \notin f(B)$; for $b \in W$, Step 1 puts $a, b$ in a connected subset of $S^2 - f(B) \subset S^2 - C$, a contradiction. So $f(\operatorname{Int} B) = V$.

**Step 3.** Open subsets of $U$ are unions of interiors of closed balls in $U$, so $f$ is open. ■

The theorem holds in $\mathbb{R}^n$, but the proof needs higher homotopy or homology groups.

## 10.3 The Jordan curve theorem

**Motivation.** The complement of a simple closed curve has exactly two components, each with boundary the curve.

**Theorem 10.8 (63.1)**

Let $X = U \cup V$ with $U, V$ open and $U \cap V = A \cup B$, $A, B$ disjoint and open. Let $\alpha$ be a path in $U$ from $a \in A$ to $b \in B$, $\beta$ a path in $V$ from $b$ to $a$, and $f = \alpha * \beta$.
(a) $[f]$ generates an infinite cyclic subgroup of $\pi_1(X, a)$.
(b) If $\pi_1(X, a)$ is infinite cyclic, it is generated by $[f]$.
(c) If $\gamma$ is a path in $U$ from $a$ to $a' \in A$ and $\delta$ a path in $V$ from $a'$ to $a$, and $g = \gamma * \delta$, then the subgroups generated by $[f]$ and $[g]$ meet only in the identity.

**Proof.**

**Step 1 (a covering space).** Let $Y \subset X \times \mathbb{Z}$ be the disjoint union of $U \times 2n$ and $V \times (2n+1)$, $n \in \mathbb{Z}$, and $E$ its quotient identifying $x \times 2n \sim x \times (2n+1)$ for $x \in B$ and $x \times 2n \sim x \times (2n-1)$ for $x \in A$; $\pi : Y \to E$ the quotient map, and $p : E \to X$ induced by $x \times m \mapsto x$. $\pi$ is open: for $W$ open in $U$, $\pi^{-1}\pi(W \times 2n) = W \times 2n \cup (W \cap B) \times (2n+1) \cup (W \cap A) \times (2n-1)$; similarly for $V$. So $p^{-1}(U)$ is the disjoint union of the open $\pi(U \times 2n)$, each mapped homeomorphically onto $U$ ($\pi | U \times 2n$ a continuous open bijection); likewise for $V$. So $p$ is a covering map.

**Step 2.** Let $e_n = \pi(a \times 2n)$, the distinct points of $p^{-1}(a)$. $\tilde{\alpha}_n(s) = \pi(\alpha(s) \times 2n)$ and $\tilde{\beta}_n(s) = \pi(\beta(s) \times (2n+1))$ lift $\alpha, \beta$; $\tilde{\alpha}_n$ ends at $\pi(b \times 2n) = \pi(b \times (2n+1))$, where $\tilde{\beta}_n$ begins, and $\tilde{\beta}_n$ ends at $\pi(a \times (2n+1)) = e_{n+1}$. So $\tilde{f}_n = \tilde{\alpha}_n * \tilde{\beta}_n$ lifts $f$ from $e_n$ to $e_{n+1}$.

**Step 3 (a).** For $m > 0$, $\tilde{f}_0 * \cdots * \tilde{f}_{m-1}$ lifts $f^m$ from $e_0$ to $e_m \neq e_0$, so $[f]^m \neq 1$.

**Step 4 (b).** The lifting correspondence $\phi$ sends $[f]^m \mapsto e_m$ ($m \in \mathbb{Z}$), so is onto. By Theorem 9.17 it induces a bijection $\pi_1(X, a)/H \to p^{-1}(a)$, $H = p_*\pi_1(E, e_0)$; the fibre is infinite while $\mathbb{Z}$ modulo a nontrivial subgroup is finite, so $H = 1$ and $\phi$ is bijective. As $\phi$ maps $\langle [f] \rangle$ onto the fibre, $\langle [f] \rangle = \pi_1(X, a)$.

**Step 5 (c).** $\tilde{\gamma}(s) = \pi(\gamma(s) \times 0)$ and $\tilde{\delta}(s) = \pi(\delta(s) \times (-1))$ lift $\gamma, \delta$, with $\pi(a' \times 0) = \pi(a' \times (-1))$ ($a' \in A$) and $\tilde{\delta}(1) = \pi(a \times (-1)) = e_0$. So every power of $g$ lifts to a loop at $e_0$, while $f^m$ ($m \neq 0$) lifts to a path ending at $e_m \neq e_0$. Hence $[f]^m = [g]^k$ forces $m = 0$ and $[g]^k = 1$. ■

**Theorem 10.9 (63.2, nonseparation theorem)**

An arc $D$ in $S^2$ does not separate $S^2$.

**Proof.** $D$ is contractible, so for $a, b \notin D$ the inclusion $D \to S^2 - a - b$ is nulhomotopic; apply Lemma 10.6. ■

A second proof gives a generalization.

**Theorem 10.10 (63.3, general nonseparation theorem)**

Let $D_1, D_2$ be closed subsets of $S^2$ with $S^2 - (D_1 \cap D_2)$ simply connected. If neither $D_1$ nor $D_2$ separates $S^2$, neither does $D_1 \cup D_2$.

**Proof.** Let $a, b \notin D_1 \cup D_2$ not be joined by a path in $S^2 - (D_1 \cup D_2)$. Put $X = S^2 - (D_1 \cap D_2)$, $U = S^2 - D_1$, $V = S^2 - D_2$; $X = U \cup V$, $U \cap V = S^2 - (D_1 \cup D_2)$. Let $A$ be the path component of $U \cap V$ containing $a$, $B$ the union of the others (both open). As $D_1, D_2$ do not separate, there are paths in $U$ from $a$ to $b$ and in $V$ from $b$ to $a$, so $\pi_1(X, a) \neq 1$ by Theorem 10.8(a), a contradiction. ■

(For an arc $D = h([0,1])$ this reproves Theorem 10.9: if $a, b$ cannot be joined in $S^2 - D$, then not in $S^2 - h(I_1)$ for a half $I_1$ of $[0,1]$, as $S^2$−point is simply connected; bisecting gives nested $I_n$ shrinking to $x$, but a path from $a$ to $b$ in $S^2 - h(x)$ misses $h(I_n)$ for large $n$.)

**Theorem 10.11 (63.4, Jordan curve theorem)**

A simple closed curve $C$ in $S^2$ separates $S^2$ into precisely two components $W_1, W_2$, and $C = \overline{W_i} - W_i$ for $i = 1, 2$.

**Proof.**

**Step 1.** Write $C = C_1 \cup C_2$, arcs meeting in $\{p, q\}$; $X = S^2 - p - q$, $U = S^2 - C_1$, $V = S^2 - C_2$, so $U \cap V = S^2 - C$ has at least two components (Theorem 10.3). Suppose more: $A_1, A_2$ two components, $B$ the union of the others, all open; $a \in A_1$, $a' \in A_2$, $b \in B$. The $C_i$ do not separate (Theorem 10.9), giving paths $\alpha, \gamma$ in $U$ from $a$ to $b$, $a'$ and $\beta, \delta$ in $V$ from $b$, $a'$ to $a$. For $f = \alpha * \beta$, $g = \gamma * \delta$, Theorem 10.8(a) for $(A_1 \cup A_2) \cup B$ and for $A_1 \cup (A_2 \cup B)$ shows $[f], [g]$ have infinite order. As $\pi_1(X, a) \cong \mathbb{Z}$, $[f]^m = [g]^k$ for some nonzero $m, k$, contradicting Theorem 10.8(c) for $(A_1 \cup A_2) \cup B$, where $a, a' \in A_1 \cup A_2$.

**Step 2.** $W_1, W_2$ are open, so $\overline{W_i} - W_i \subset C$. Conversely let $x \in C$, $U \ni x$ a neighbourhood; write $C = C_1 \cup C_2$, arcs meeting in their end points, with $C_1 \subset U$. For $a \in W_1$, $b \in W_2$ take a path $\alpha$ from $a$ to $b$ in $S^2 - C_2$ (Theorem 10.9). Its connected image meets the disjoint open $W_1$ and $S^2 - \overline{W_1} \supset W_2$, so contains a point $y \in \overline{W_1} - W_1 \subset C$, and $y \in C_1 \subset U$. So $x \in \overline{W_1} - W_1$; likewise for $W_2$. ■

**Theorem 10.12 (63.5)**

If $C_1, C_2$ are closed connected subsets of $S^2$ meeting in exactly two points, and neither separates $S^2$, then $C_1 \cup C_2$ separates $S^2$ into precisely two components.

**Proof.** At least two by Theorem 10.4; at most two by Step 1 above, which used only these properties. ■

Two arcs meeting in two points not both end points separate $S^2$ into two components, but need not be their common boundary.

## 10.4 Imbedding graphs in the plane

**Motivation.** The Jordan curve theorem shows $K_{3,3}$ (three houses, three utilities) is not planar.

A **linear graph** is a Hausdorff space that is a finite union of arcs (**edges**), any two meeting in at most a common end point (a **vertex**). The **complete graph on $n$ vertices** has an edge joining each pair of vertices; the **utilities graph** has vertices $g, w, e, h_1, h_2, h_3$ with an edge joining each of $g, w, e$ to each $h_i$. A **theta space** is a Hausdorff space that is the union of three arcs $A, B, C$, each pair meeting precisely in their common end points.

**Lemma 10.13 (64.1)**

A theta space $X = A \cup B \cup C$ in $S^2$ separates $S^2$ into three components, with boundaries $A \cup B$, $B \cup C$, $A \cup C$. The component with boundary $A \cup B$ is a component of $S^2 - (A \cup B)$.

**Proof.** Let $a, b$ be the common end points. $A \cup B$ separates $S^2$ into $U, U'$ with boundary $A \cup B$ (Jordan curve theorem), and the connected $C - a - b$ lies in one, say $U'$. The connected $\overline{U} = U \cup A \cup B$ and $C$ meet only in $\{a, b\}$ and neither separates ($C$ is an arc, $S^2 - \overline{U} = U'$). By Theorem 10.12, $\overline{U} \cup C$ separates $S^2$ into $V, W$. So $S^2 - X = U \cup V \cup W$, disjoint open connected sets, its components; $U$ has boundary $A \cup B$ and is a component of $S^2 - (A \cup B)$. By symmetry the others have boundaries $B \cup C$, $A \cup C$. ■

**Theorem 10.14 (64.2)**

The utilities graph cannot be imbedded in the plane.

**Proof.** Suppose it lies in $S^2$. $A = gh_1 \cup h_1 w$, $B = gh_2 \cup h_2 w$, $C = gh_3 \cup h_3 w$ form a theta space, separating $S^2$ into $U, V, W$ with boundaries $A \cup B$, $B \cup C$, $A \cup C$. $e$ and the connected $eh_i - h_i$ lie in one of them, so $h_1, h_2, h_3$ lie in its boundary. But $A \cup B \not\ni h_3$, $B \cup C \not\ni h_1$, $A \cup C \not\ni h_2$. ■

**Lemma 10.15 (64.3)**

A complete graph $X$ on four vertices $a_1, \ldots, a_4$ in $S^2$ separates $S^2$ into four components, whose boundaries are the subgraphs $X_i$ consisting of the edges not having $a_i$ as a vertex.

**Proof.** The union $Y$ of all edges but $a_2a_4$ is the theta space with arcs $A = a_1a_2 \cup a_2a_3$, $B = a_1a_3$, $C = a_1a_4 \cup a_4a_3$, separating $S^2$ into $U, V, W$ with boundaries $A \cup B$, $B \cup C$, $A \cup C$. The connected $a_2a_4 - a_2 - a_4$ lies in one whose boundary contains $a_2, a_4$: not $U$ ($a_4 \notin A \cup B$), not $V$ ($a_2 \notin B \cup C$), so $W$. $\overline{U} \cup \overline{V}$ is connected (both contain $B$) and does not separate (complement $W$), the arc $a_2a_4$ does not separate, and they meet only in $a_2, a_4$; by Theorem 10.12 their union separates $S^2$ into $W_1, W_2$. So $S^2 - X = U \cup V \cup W_1 \cup W_2$, its components. $U$ has boundary $A \cup B = X_4$; by symmetry the others have $X_1, X_2, X_3$. ■

**Theorem 10.16 (64.4)**

The complete graph on five vertices cannot be imbedded in the plane.

**Proof.** Suppose $G \subset S^2$ has vertices $a_1, \ldots, a_5$; let $X$ be the complete subgraph on $a_1, \ldots, a_4$. $a_5$ and the connected set of edges at $a_5$ minus $a_1, \ldots, a_4$ lie in one component of $S^2 - X$, so $a_1, \ldots, a_4$ lie in its boundary $X_i$; but $a_i \notin X_i$. ■

By Kuratowski's theorem, a graph imbeds in the plane iff it contains no subgraph homeomorphic to the utilities graph or the complete graph on five vertices.

## 10.5 Winding number of simple closed curves

**Motivation.** The winding number of a simple closed curve is $\pm 1$ inside and 0 outside.

**Lemma 10.17 (65.1)**

Let $G \subset S^2$ be a complete graph on $a_1, \ldots, a_4$, $C$ the simple closed curve $a_1a_2a_3a_4a_1$, and $p, q$ interior points of the edges $a_1a_3$ and $a_2a_4$. Then (a) $p$ and $q$ lie in different components of $S^2 - C$;
(b) the inclusion $j : C \to S^2 - p - q$ induces an isomorphism of fundamental groups.

**Proof.** (a) The theta space $C \cup a_1a_3$ (arcs $a_1a_2a_3$, $a_1a_3$, $a_1a_4a_3$) separates $S^2$ into three components; one, $W$, has boundary $C$, the only one whose boundary contains $a_2$ and $a_4$. So $a_2a_4 - a_2 - a_4 \subset W$ and $q \in W$. By Lemma 10.13, $W$ is a component of $S^2 - C$, and $p \notin W$ ($p$ is on the theta space).

(b) Let $X = S^2 - p - q$, $x$ interior to $a_1a_2$, $y$ interior to $a_3a_4$, $\alpha$ the path in $C$ from $x$ through $a_2, a_3$ to $y$, $\beta$ the path in $C$ from $y$ through $a_4, a_1$ to $x$. Let $D_1$, $D_2$ be the arcs of $D = a_1a_3a_2a_4a_1$ from $p$ through $a_1, a_4$, resp. $a_3, a_2$, to $q$; $U = S^2 - D_1$, $V = S^2 - D_2$. Then $X = U \cup V$, $U \cap V = S^2 - D$ has two components (Jordan curve theorem), containing $x$ and $y$ respectively by (a) for $G$ with the curve $D$ ("diagonals" $a_1a_2$, $a_3a_4$); $\alpha \subset U$, $\beta \subset V$. As $\pi_1(X, x) \cong \mathbb{Z}$, $[\alpha * \beta]$ generates it (Theorem 10.8(b)). Since $\alpha * \beta$ is a loop in $C$, $j_* : \pi_1(C, x) \cong \mathbb{Z} \to \pi_1(X, x) \cong \mathbb{Z}$ is onto, hence an isomorphism. ■

**Theorem 10.18 (65.2)**

If $C$ is a simple closed curve in $S^2$ and $p, q$ lie in different components of $S^2 - C$, then $j : C \to S^2 - p - q$ induces an isomorphism of fundamental groups.

**Proof.**

**Step 1.** If $A$ is an arc from $a$ to $b$ and $B$ one from $b$ to $c$ ($a, b, c$ distinct), $A \cup B$ contains an arc from $a$ to $c$: for homeomorphisms $f : I \to A$, $g : I \to B$ (from $a$ to $b$, $b$ to $c$), let $t_0$ be least with $f(t_0) \in B$, $g(t_1) = f(t_0)$; take $f([0, t_0]) \cup g([t_1, 1])$.

**Step 2.** Any two points of a connected open $U \subset \mathbb{R}^2$ are end points of an arc in $U$: "equal, or joined by an arc in $U$" is an equivalence relation (Step 1) with open classes (segments in a disc).

**Step 3.** A complete graph on four vertices containing $C$ as $a_1a_2a_3a_4a_1$. In $\mathbb{R}^2$, with 0 in the bounded component of $\mathbb{R}^2 - C$, let $a_1$ be the largest point of $C$ on the negative $x$-axis, $a_3$ the smallest on the positive one; the open segment between them misses $C$ and contains 0, so lies in the bounded component. Write $C = C_1 \cup C_2$, arcs from $a_1$ to $a_3$, and take $a$ in the unbounded component. As $C_1, C_2$ do not separate, Step 2 gives injective paths $\alpha$ in $\mathbb{R}^2 - C_1$, $\beta$ in $\mathbb{R}^2 - C_2$ from $a$ to 0. Let $t_0$ be least with $\alpha(t_0) \in C$, $a_2 = \alpha(t_0)$, interior to $C_2$; similarly $a_4 = \beta(t_1)$ interior to $C_1$. $\alpha([0, t_0])$, $\beta([0, t_1])$ meet $C$ only at their end points and otherwise lie in the unbounded component; by Step 1 their union contains an arc from $a_2$ to $a_4$ meeting $C \cup a_1a_3$ only in $a_2, a_4$. With $a_1a_3$ and $C$ it forms the graph, so by Lemma 10.17 the theorem holds for some $p, q$ in different components.

**Step 4.** Moving $p$ (and symmetrically $q$) in its component. With $q = \infty$: let $D$ be a simple closed curve, $0, p$ in one component of $\mathbb{R}^2 - D$, and $j : D \to \mathbb{R}^2 - 0$ inducing an isomorphism. For $f(x) = x - p$, a homeomorphism $\mathbb{R}^2 - p \to \mathbb{R}^2 - 0$, and a path $\gamma$ in $\mathbb{R}^2 - D$ from 0 to $p$, $F(x, t) = x - \gamma(t)$ is a homotopy in $\mathbb{R}^2 - 0$ from $j$ to $f \circ k$, $k : D \to \mathbb{R}^2 - p$ the inclusion. By Corollary 9.35, $(f \circ k)_*$ and hence $k_*$ are isomorphisms. ■

## 10.6 The Cauchy integral formula

**Motivation.** The facts above justify the Cauchy integral formula for simple closed curves.

For a loop $f$ in $\mathbb{R}^2$ and $a \notin f(I)$, let $g(s) = (f(s) - a)/\|f(s) - a\|$, a loop in $S^1$, and $\tilde{g}$ a lifting to $\mathbb{R}$. The integer $n(f, a) = \tilde{g}(1) - \tilde{g}(0)$ is the **winding number** of $f$ about $a$, independent of the lifting (two differ by a constant). A **free homotopy** of loops is a map $F : I \times I \to X$ with $F(0, t) = F(1, t)$ for all $t$.

**Lemma 10.19 (66.1)**

Let $f$ be a loop in $\mathbb{R}^2 - a$.
(a) $n(\bar{f}, a) = -n(f, a)$.
(b) If $f$ is freely homotopic to $f'$ through loops in $\mathbb{R}^2 - a$, then $n(f, a) = n(f', a)$.
(c) If $a, b$ lie in the same component of $\mathbb{R}^2 - f(I)$, then $n(f, a) = n(f, b)$.

**Proof.** (a) $s \mapsto 1 - s$ reverses the sign of $\tilde{g}(1) - \tilde{g}(0)$.

(b) For a free homotopy $F$, lift $G = (F - a)/\|F - a\|$ to $\tilde{G} : I \times I \to \mathbb{R}$ (Lemma 9.13); $\tilde{G}(1, t) - \tilde{G}(0, t)$ is continuous and integer valued, hence constant.

(c) $n(f, a) = n(f - a, 0)$, and for a path $\gamma$ in $\mathbb{R}^2 - f(I)$ from $a$ to $b$, $f(s) - \gamma(t)$ is a free homotopy in $\mathbb{R}^2 - 0$ from $f - a$ to $f - b$. ■

A loop $f$ is **simple** if $f(s) = f(s')$ only when $s = s'$ or $\{s, s'\} = \{0, 1\}$; its image is a simple closed curve.

**Theorem 10.20 (66.2)**

Let $f$ be a simple loop in $\mathbb{R}^2$. If $a$ is in the unbounded component of $\mathbb{R}^2 - f(I)$, $n(f,a) = 0$; if $a$ is in the bounded component, $n(f,a) = \pm 1$.

**Proof.** Take $a = 0$ and, rotating (a free homotopy), the base point $x_0$ on the positive $x$-axis. Let $C = f(I)$, $h : S^1 \to C$ with $h \circ p_0 = f$; $h_*[p_0]$ generates $\pi_1(C, x_0)$. If 0 is in the bounded component, $[f] = j_* h_*[p_0]$ generates $\pi_1(\mathbb{R}^2 - 0, x_0)$ (Theorem 10.18, $\infty$ in the other component); if in the unbounded one, $j \circ h$ is nulhomotopic (Lemma 10.2) and $[f] = 1$. The retraction $x \mapsto x/\|x\|$ induces an isomorphism onto $\pi_1(S^1, b_0)$, so $g = f/\|f\|$ represents a generator or the identity: its lifting from 0 ends at $\pm 1$ or 0 (proof of Theorem 9.16). ■

A simple loop is **counterclockwise** if $n(f,a) = +1$ for $a$ in the bounded component and **clockwise** if it is $-1$.

**Lemma 10.21 (66.3)**

If $f$ is a piecewise-differentiable loop in $\mathbb{C}$ and $a \notin f(I)$, then $n(f,a) = \dfrac{1}{2\pi i} \displaystyle\int_f \frac{dz}{z-a}$.

**Proof.** Let $r = |f - a|$, $g = (f-a)/r$, $\tilde{g}$ a lifting, $\theta = 2\pi\tilde{g}$; then $f(s) - a = r(s)e^{i\theta(s)}$, $r, \theta$ piecewise differentiable ($\theta$ locally a branch of the argument), and

$$\int_f \frac{dz}{z-a} = \int_0^1 \frac{f'(s)}{f(s)-a}\,ds = \int_0^1 \Big(\frac{r'(s)}{r(s)} + i\theta'(s)\Big)ds = [\log r]_0^1 + i[\theta]_0^1 = 2\pi i\, n(f,a).$$

■

**Theorem 10.22 (66.4, Cauchy integral formula, classical version)**

Let $C$ be a simple closed piecewise-differentiable curve in $\mathbb{C}$, $B$ the bounded component of $\mathbb{R}^2 - C$, and $F$ analytic on an open set $\Omega \supset B \cup C$. For $a \in B$,

$$F(a) = \pm \frac{1}{2\pi i} \int_C \frac{F(z)}{z-a}\,dz,$$

with $+$ if $C$ is oriented counterclockwise and $-$ otherwise.

**Proof.** General version (Ahlfors): if $F$ is analytic on a region $\Omega$, $f$ a piecewise-differentiable loop in $\Omega$ with $n(f,b) = 0$ for all $b \notin \Omega$, and $a \in \Omega - f(I)$, then $n(f,a)F(a) = \frac{1}{2\pi i}\int_f \frac{F(z)}{z-a}dz$. For a parametrization $f$ of $C$, every $b \notin \Omega$ is in the unbounded component, so $n(f,b) = 0$, and $n(f,a) = \pm 1$ for $a \in B$ (Theorem 10.20). ■